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Shortcuts and Temporal Leakage in the Bangalore EEG Epilepsy Dataset

A Provenance Audit and Leakage-Aware Benchmark of Lightweight Seizure-Type Classifiers

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1. Introduction: Motivation

- Epilepsy affects about 50 million people worldwide [4].
- Visual EEG review is expert-intensive and slow.
- Automated seizure-type classification could support clinicians.
- Public benchmarks like BEED [1] drive models; their validity matters.

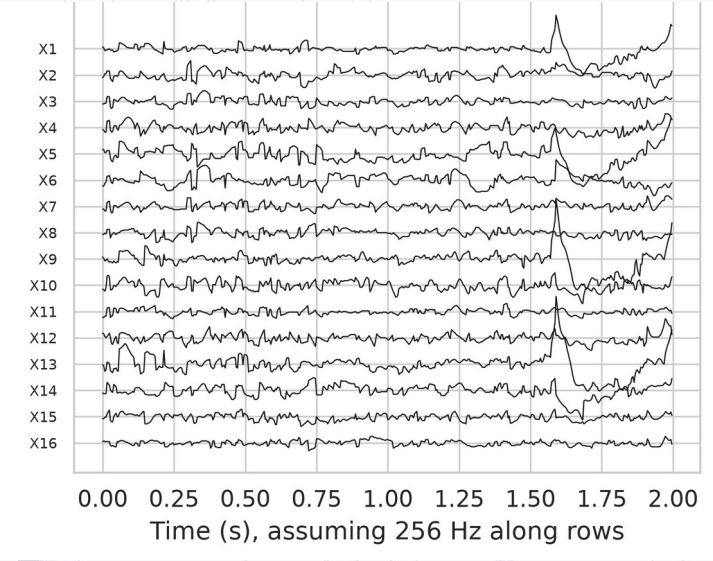


Fig. 1. BEED focal-seizure class: 512 consecutive rows drawn as 16 traces [1].

1. Introduction: The BEED Dataset

8,000 rows · 16 values per row (X1–X16) · 256 Hz, 16 channels [1]

0 · Healthy

2,000 rows; documented as healthy adult subjects [1].

1 · Generalised seizures

2,000 rows; seizures involving both hemispheres.

2 · Focal seizures

2,000 rows; onset in a restricted brain region.

3 · Seizure events

2,000 rows; events with blinking or staring behaviours.

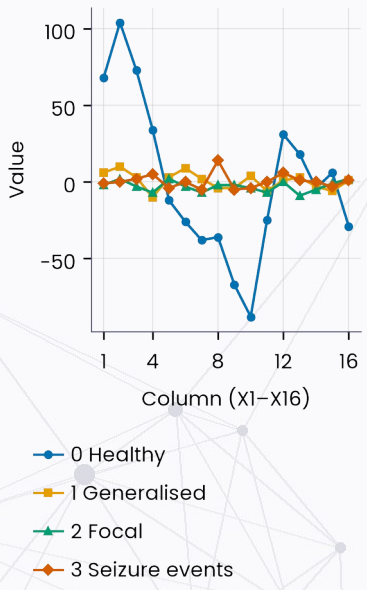


Fig. 2. One BEED row per class; healthy rows show far larger amplitude.

1. Introduction: Research Question

How well do models classify BEED seizure types when tested without leakage?

Sub-question: the origin of the healthy rows.

Random row split

Standard practice in prior BEED studies.

Adjacent rows can land in train and test.

Leakage-safe block split

200-row blocks with 50-row gaps.

No near-duplicates across the split.

2. Dataset Audit: Where the Healthy Class Comes From

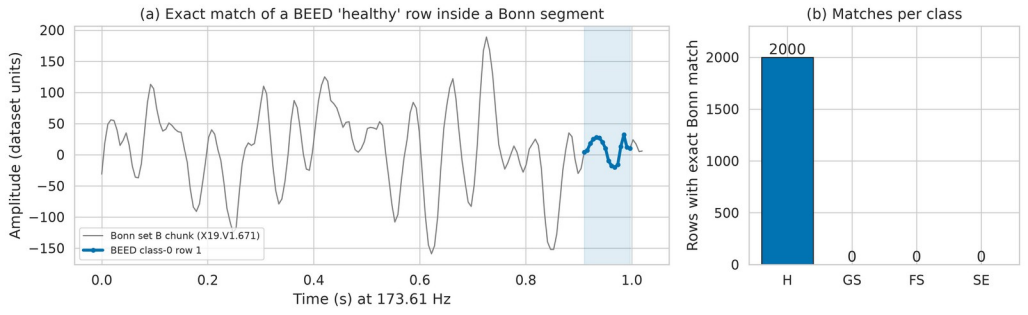


Fig. 3. A BEED healthy row identical to a Bonn set-B window (a); matches per class (b).

Match — All 2,000 healthy rows match UCI data exactly [3]; classes 1–3 do not.

Source — Bonn set B, eyes closed, 173.61 Hz [2]; classes 1–3: 256 Hz.

Position — Every match lies at samples 159–174 of a one-second chunk.

Effect — A single smoothness threshold separates healthy rows (accuracy 0.997).

2. Dataset Audit: Temporal Leakage

Consecutive seizure-class rows are adjacent time points, hence near-duplicates.

Random split 88–95% of seizure test rows have their neighbouring row in training.

Implication Random-split scores measure interpolation, not generalisation.

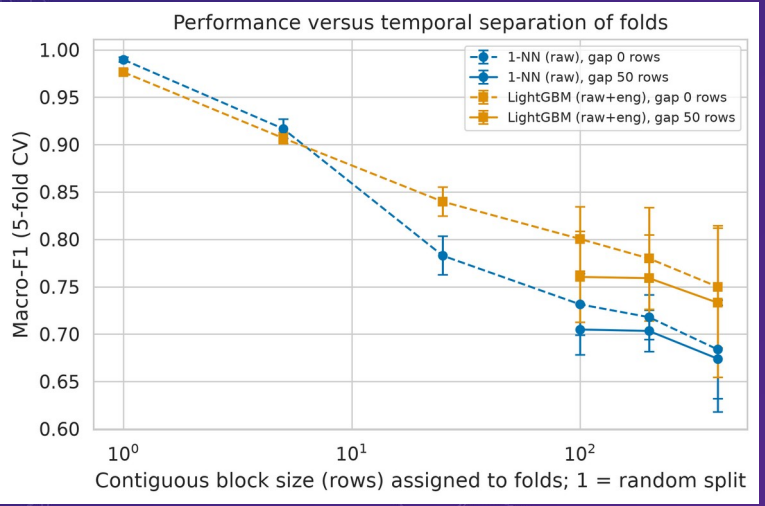


Fig. 4. Macro-F1 falls as folds use longer row blocks.

3. Methodology

Nested tuning on macro-F1; each test set used once

Splits

Random 80/20 vs blocked hold-out: 200-row blocks, 50-row purge; 5×2 CV.

Models

LR, kNN, SVM-RBF, RF, XGBoost, LightGBM; MLP, 1D-CNN, BiLSTM, Transformer.

Statistics

Cluster-bootstrap 95% CIs; corrected resampled t-test; Friedman + Nemenyi.

Metrics

Macro-F1 (primary), accuracy, Cohen's kappa, AUC, calibration (ECE).

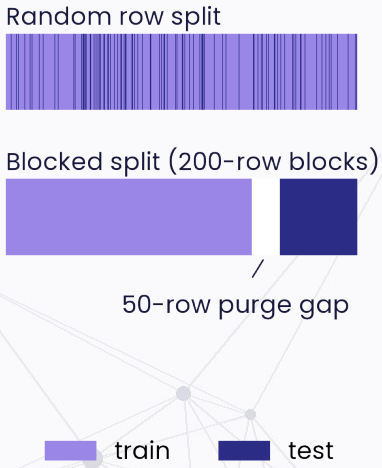


Fig. 5. Schematic: random versus blocked hold-out with a purge gap.

4. Results: Random vs Leakage-Safe Split

	Random	Leakage-safe
4-class macro-F1	0.986	0.842
95% CI	0.980–0.992	0.800–0.880
Accuracy	0.986	0.844
Cohen's kappa	0.982	0.792
3-class macro-F1	0.990	0.776
Median ECE	0.029	0.093

Best models: kNN (random), SVM-RBF (leakage-safe); raw features. 3-class: seizure types only.

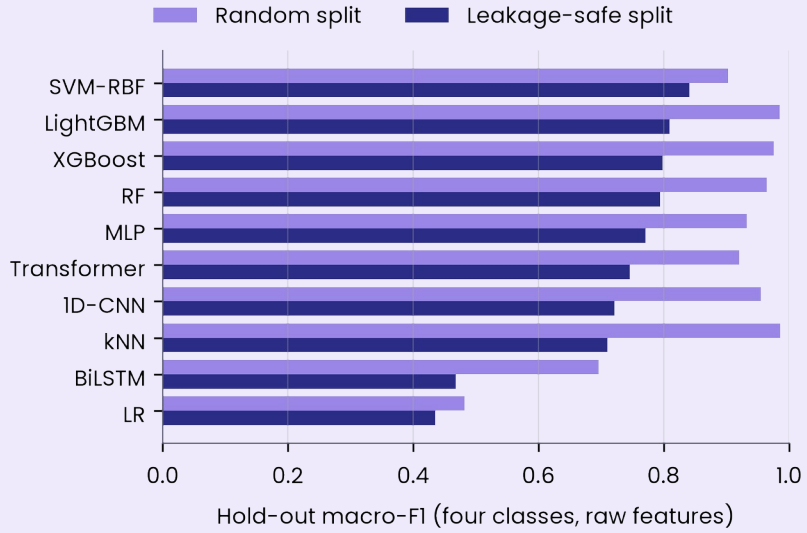


Fig. 6. Four-class hold-out macro-F1 per model, raw features, by split.

4. Results: What the Models Look At

Leakage-safe feature importance

- Seizure types: raw voltages of channels X11, X8 and X15.
- Healthy class: smoothness, e.g. Hjorth mobility.

Healthy class: a recording-source shortcut

- Healthy vs seizure: macro-F1 ≥ 0.999 under both splits.
- One lag-1 autocorrelation threshold: accuracy 0.997.

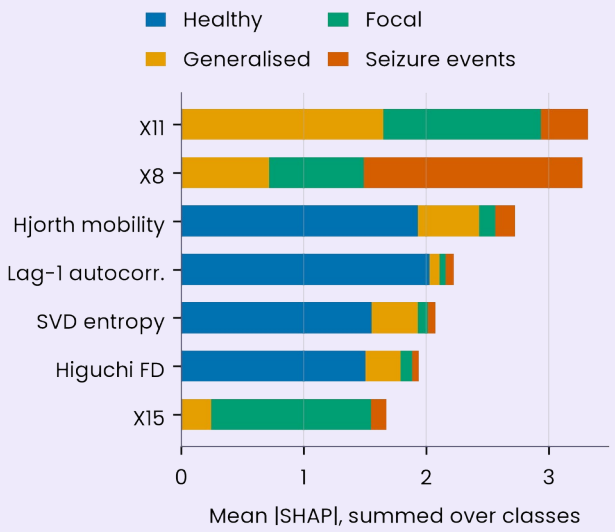


Fig. 7. Top SHAP features, LightGBM, four classes, leakage-safe split.

4. Results: Lightweight Edge Model

Model (raw 16 values)	Parameters	Size (kB)	Latency (ms)	Macro-F1*
MLP	**11,076**	**43**	**0.077**	**0.770**
1D-CNN	19,524	76	0.145	0.714
Transformer	17,796	70	0.249	0.745
LightGBM (raw+eng.)	62,720	≈735	4.350	0.794

**Leakage-safe hold-out. Latency: median of 500 single-sample runs, one Xeon vCPU thread.*

- The MLP is far below the 3.9 ms sampling interval at 256 Hz: suited to edge devices.
- Best leakage-safe hold-out model, SVM-RBF (0.842), was not profiled.
- 5×2 blocked CV: best LightGBM 0.767 ± 0.057; Friedman test $p < 0.001$.

5. Discussion: Limitations

What remains valid

The blocked benchmark is a lower-leakage estimate of generalisation to unseen time within the same recordings.

Data

- No subject IDs: patient-independent testing is impossible.
- Recording and class are confounded.
- Healthy class from another source, sampled at 173.61 Hz.

Scope

- Classes 1–3 not checked against other public databases.
- Single dataset; no external test.

6. Conclusion: Takeaways and Future Work

Takeaways

- Healthy rows identical to Bonn set B windows [2], [3]: a source shortcut.
- Random splits leak: macro-F1 0.986 vs 0.842 (3-class 0.990 vs 0.776).
- Report blocked splits as the BEED benchmark.
- An 11k-parameter MLP (0.077 ms per sample) is edge-ready.

Future work

- Subject IDs for patient-independent tests.
- Same-source healthy recordings.
- Cross-dataset validation on public EEG sets.
- Clinical and wearable deployment studies.

References

[1] Najmusseher N, Nizar Banu PK. BEED: Bangalore EEG Epilepsy Dataset [dataset]. UCI Machine Learning Repository. 2024. doi:10.24432/C5K33B.

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[3] Wu Q, Fokoue E. Epileptic Seizure Recognition data set [dataset]. UCI Machine Learning Repository. 2017. Copy distributed by the creators at github.com/QiuyiWu/Epileptic-Seizure-Recognition-Data (accessed 7 October 2026).

[4] World Health Organization. Epilepsy [fact sheet]. who.int/news-room/fact-sheets/detail/epilepsy (accessed 8 October 2026).

Complete reference list: see the full paper.

THANK YOU

FOR YOUR ATTENTION

Questions?

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